Dorey's Resource Book

Topic 2 – Relative frequency and probability

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1 The probability scale

ANCHOR QUESTION

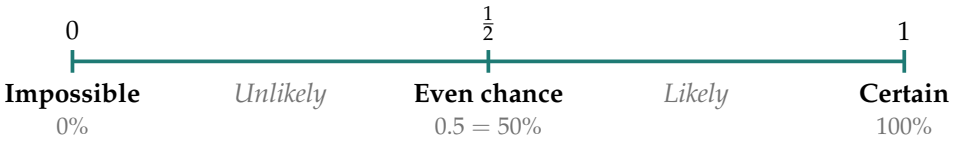
The probability that it rains tomorrow is 0.8. *Write this as a percentage and describe the chance in words.*

SYLLABUS MST-12-S1-07

- Recognise that probabilities range from 0 (impossible) to 1 (certain) and can be represented as fractions, decimals and percentages

NOTES

- **Probability** measures how likely an event is to happen.
- Every probability is between 0 and 1, including 0 and 1. As a percentage, it is between 0% and 100%.
- 0 means **impossible**, $\frac{1}{2}$ means **even chance** and 1 means **certain**.
- Between 0 and $\frac{1}{2}$ is **unlikely**. Between $\frac{1}{2}$ and 1 is **likely**.
- A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$.
- To compare probabilities, write them all in the same form first.



WORKED EXAMPLE

The probability that a randomly chosen phone battery lasts more than two days is 0.25. *Write this as a percentage and describe the chance in words.*

$0.25 \times 100 = 25\%$ — decimal to percentage: $\times 100$
25% is less than 50% — compare with even chance
The chance is **unlikely**. — between 0 and $\frac{1}{2}$

Tier 1 · Decimal to percentage and words

1. Which of these numbers could be a probability?

$$0.4 \qquad 1.3 \qquad 75\% \qquad -0.1 \qquad 120\% \qquad \frac{2}{3}$$

2. Describe each probability in words.
(a) 0 (b) $\frac{1}{2}$ (c) 0.7 (d) 100%

3. The probability that a train arrives on time is 0.9. Write this as a percentage and describe the chance in words.

4. The probability that a new light globe is faulty is 0.02. Write this as a percentage and describe the chance in words.

Tier 2 · Starting from a fraction

5. The probability that a randomly chosen student walks to school is $\frac{2}{5}$. Write this as a decimal and as a percentage, and describe the chance in words.

6. The probability that a flight leaves on time is $\frac{17}{20}$. Write this as a decimal and as a percentage, and describe the chance in words.

7. The probability of winning a prize in a game is $\frac{1}{8}$. Write this as a decimal and as a percentage, and describe the chance in words.

Tier 3 · Starting from a percentage

8. A forecast gives a 60% chance of rain. Write this as a fraction in simplest form and as a decimal, and describe the chance in words.

9. In a raffle, 4% of the tickets win a prize. Write the probability that a ticket wins as a fraction in simplest form and as a decimal, and describe the chance in words.

10. A basketballer makes 72% of her free throws. Write the probability that she makes her next free throw as a fraction in simplest form and as a decimal, and describe the chance in words.

Tier 4 · Comparing and ordering

11. Order these probabilities from least likely to most likely: 0.3, $\frac{2}{5}$, 35%.

12. Order these probabilities from least likely to most likely: $\frac{7}{8}$, 0.85, 88%.

13. The probabilities of three teams winning a competition are: Hawks 0.45, Lions $\frac{9}{20}$, Owls 44%. Which team is least likely to win? Are any two teams equally likely to win?

Tier 5 · Judging from the situation

14. Use one of the words *impossible*, *unlikely*, *even chance*, *likely* or *certain* to describe each event.

(a) A fair coin lands on heads.

(b) A standard six-sided die is rolled and shows a 7.

(c) A randomly chosen month has at least 28 days.

15. One student from a class of 30 is chosen at random to be class captain. Which of the following best describes the probability that Ella, a student in the class, is chosen?

A. Certain

B. Even chance

C. Unlikely

D. Impossible

16. A bag holds 9 green counters and 1 red counter. A counter is taken out without looking. Describe, in words, the chance that it is green.

Conceptual

17. Explain why 1.4 cannot be a probability.

18. Sam says a probability of 0.08 is more likely than a probability of 0.5 because 8 is bigger than 5. Explain his mistake.

19. An event has an even chance of happening. Write its probability as a fraction, a decimal and a percentage.

20. A weather app gives a 0% chance of rain for tomorrow. Explain what this means, and why it might still rain.

21. The probability that a phone made at Factory A is faulty is 0.03.

(a) Write this probability as a percentage. (1 mark)

(b) Describe this chance in words. (1 mark)

(c) The probability that a phone made at Factory B is faulty is $\frac{1}{25}$. Which factory is more likely to make a faulty phone? Justify your answer. (2 marks)

Revision

22. A fruit punch is made by mixing juice and soda water in the ratio 3 : 2.

(a) What fraction of the punch is soda water? (1 mark)

(b) How many litres of juice are needed to make 4 L of punch? (2 marks)

(c) Juice costs \$3.60 per 2 L bottle. Find the cost of the juice for 4 L of punch. (2 marks)

23. Kira drives 342 km in 3 hours 48 minutes.

(a) Write 3 hours 48 minutes in hours, as a decimal. (1 mark)

(b) Find Kira's average speed in km/h. (1 mark)

(c) Convert this speed to m/s. (1 mark)

(d) Her car uses 7.5 L/100 km and petrol costs \$1.96 per litre. Find the cost of the fuel for the trip, to the nearest cent. (2 marks)

24. The number of goals scored by a netball team in eight games is shown.

4 7 7 8 9 10 12 15

(a) Find the mean. (1 mark)

(b) Find the median. (1 mark)

(c) Find the range. (1 mark)

(d) Find the interquartile range. (2 marks)

2 Sample spaces

ANCHOR QUESTION

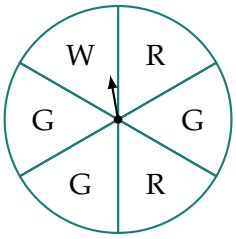
A standard six-sided die is rolled. List the sample space and state the number of outcomes.

SYLLABUS MST-12-S1-07

- Identify and define the sample space as the set of all possible outcomes of a chance experiment
- Determine the number of outcomes for an experiment

NOTES

- A **chance experiment** is an action with a result that cannot be predicted, such as tossing a coin or rolling a die.
- Each possible result is an **outcome**.
- The **sample space**, S, is the set of all possible outcomes, written in braces: tossing a coin gives $S = \{H, T\}$.
- List every possible outcome, each one **only once**.
- The number of outcomes is not always the number of sections: the spinner below has 6 sections but $S = \{R, G, W\}$, so 3 outcomes.



WORKED EXAMPLE

A spinner has 5 equal sections numbered 1 to 5. List the sample space and state the number of outcomes.

$S = \{1, 2, 3, 4, 5\}$ — every possible result, each once

5 outcomes — count the outcomes in S

Tier 1 · Listing the sample space

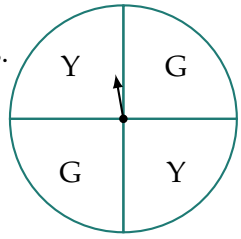
25. A coin is tossed. List the sample space and state the number of outcomes.

26. A spinner has 3 equal sections labelled A, B and C. List the sample space and state the number of outcomes.

27. A day of the week is chosen at random. List the sample space and state the number of outcomes.

Tier 2 · Repeated sections

28. List the sample space for this spinner and state the number of outcomes.



29. A bag holds 5 red, 3 blue and 2 white marbles. One marble is taken out without looking. List the sample space and state the number of outcomes.

30. A spinner has 6 equal sections numbered 1, 1, 2, 2, 2, 3. List the sample space and state the number of outcomes.

Tier 3 · Letters in a word

31. A letter is chosen at random from the word MISSISSIPPI. List the sample space and state the number of outcomes.

32. A letter is chosen at random from the word BALLOON. List the sample space and state the number of outcomes.

33. A letter is chosen at random from the word STATISTICS. List the sample space and state the number of outcomes.

Tier 4 · Counting without listing

34. A card is drawn from a standard deck of 52 playing cards. How many outcomes are in the sample space?

35. One student is chosen at random from a class of 28. How many outcomes are in the sample space?

36. Raffle tickets are numbered 100 to 499. One ticket is drawn. How many outcomes are in the sample space?

Tier 5 · Outcomes for an event

37. A standard die is rolled. List the outcomes that are greater than 4.

38. A number from 1 to 30 is chosen at random. List the outcomes that are multiples of 7.

39. A number from 1 to 20 is chosen at random. List the odd numbers greater than 12.

Conceptual

40. A spinner has 6 equal segments: 2 are red, 3 are green and 1 is white. Explain why its sample space is $S = \{R, G, W\}$ and not $S = \{R, R, G, G, G, W\}$.

41. Jo says the sample space for choosing a letter from the word BOOK is $\{B, O, O, K\}$, so there are 4 outcomes. Explain her mistake.

42. The sample space for a spinner is $S = \{1, 2, 3, 4, 5\}$. Could the spinner have 8 sections? Explain.

43. A soccer team plays a match. Write the sample space for the result of the match and state the number of outcomes.

44. A wheel has the numbers 1 to 30 marked on it. When the wheel is spun, each number is equally likely to win.

(a) How many outcomes are in the sample space? (1 mark)

(b) List all the even numbers on the wheel that are greater than 21. (1 mark)

(c) How many of the outcomes are NOT even numbers greater than 21? (1 mark)

Revision

45. The probability that a randomly chosen car at a service station is electric is $\frac{3}{40}$.

(a) Write this probability as a decimal. (1 mark)

(b) Write this probability as a percentage. (1 mark)

(c) Describe this chance in words. (1 mark)

(d) The probability that a randomly chosen car is a ute is 12%. Is a car more likely to be electric or a ute? Justify your answer. (2 marks)

46. Mia earns \$28.40 per hour for a 38-hour week. Overtime is paid at time-and-a-half.

(a) Find Mia's pay for 38 normal hours. (1 mark)

(b) Find Mia's overtime rate of pay. (1 mark)

(c) One week, Mia works 38 normal hours and 4 hours of overtime. Find her total pay for the week. (2 marks)

(d) Mia works the same hours the next week. Find her pay for the fortnight. (1 mark)

47. Rice is sold in a 1 kg bag for \$3.20 or a 5 kg bag for \$14.50.

(a) Find the price per kilogram of the 5 kg bag. (1 mark)

(b) Which bag is the better buy? (1 mark)

(c) How much is saved by buying one 5 kg bag instead of five 1 kg bags? (1 mark)

(d) A recipe uses 150 g of rice per serve. How many full serves can be made from the 5 kg bag? (2 marks)

3 Equally likely outcomes

ANCHOR QUESTION

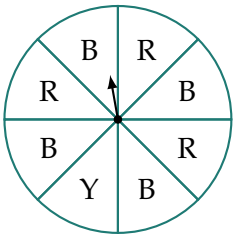
A standard die is rolled. Find P(an even number).

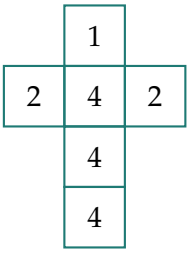
SYLLABUS MST-12-S1-07

- Express the probability of an event that has a finite number of equally likely outcomes, as $P(\text{event}) = \dfrac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

NOTES

- Outcomes are **equally likely** when each one has the same chance of happening, such as the faces of a fair die.
- When the outcomes are equally likely, $P(\text{event}) = \dfrac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$.
- P(even) is read "the probability of an even number". Simplify the fraction, or write it as a decimal or percentage if asked.
- On a spinner, count the **equal sections**, not the colours: 3 red sections out of 8 gives $P(\text{red}) = \frac{3}{8}$.
- Repeated letters or faces count every time they appear.





WORKED EXAMPLE

A standard die is rolled. Find P(a number greater than 4).

Favourable outcomes: 5, 6 — list the outcomes that fit

$P = \dfrac{2}{6}$ — favourable ÷ total

$= \dfrac{1}{3}$ — simplify

Tier 1 · Dice and coins

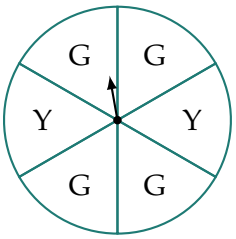
48. A fair coin is tossed. Find P(head).

49. A standard die is rolled. Find P(an odd number).

50. A standard die is rolled. Find P(a number less than 5).

Tier 2 · Spinners with repeated sections

51. For this spinner, find $P(\text{green})$.



52. A spinner has 10 equal sections: 5 red, 3 blue and 2 white. Find $P(\text{blue})$.

53. A spinner has 12 equal sections: 6 are numbered 1, 4 are numbered 2 and 2 are numbered 3. Find $P(3)$.

Tier 3 · Finding the total first

54. A bag holds 4 red and 6 green counters. One is taken out at random. Find $P(\text{red})$.

55. A jar holds 8 strawberry, 5 lime and 7 orange lollies. One is chosen at random. Find $P(\text{lime})$.

56. A class has 12 boys and 18 girls. One student is chosen at random. Find the probability that a girl is chosen, as a percentage.

Tier 4 · Letters in a word

57. A letter is chosen at random from the word BANANA. Find $P(\text{A})$.

58. A letter is chosen at random from the word STATISTICS. Find $P(\text{T})$.

59. A letter is chosen at random from the word MATHEMATICS. Find $P(\text{a vowel})$.

Tier 5 · Two conditions

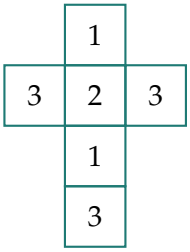
60. A number from 1 to 10 is chosen at random. Find $P(\text{an even number greater than } 5)$.

61. A number from 1 to 30 is chosen at random. Find the probability that it is a multiple of 4 or a multiple of 6.

62. A number from 1 to 50 is chosen at random. Find the probability that it is an odd multiple of 5, as a decimal.

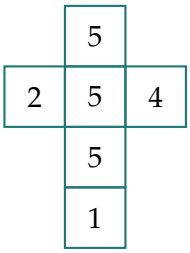
Tier 6 · Nets and biased dice

63. A die is made from this net and rolled once. Find $P(3)$.



64. A die has faces 1, 1, 1, 1, 2, 6. It is rolled once. Find $P(1)$.

65. A die is made from this net and rolled once. Which is more likely: rolling a 5 or rolling an even number?



Conceptual

66. A spinner has a red half, a blue quarter and a green quarter. Lily says $P(\text{red}) = \frac{1}{3}$ because there are three colours. Explain her mistake and find the correct probability.

67. A footy match can end in a win, a draw or a loss. Explain why $P(\text{draw})$ is not $\frac{1}{3}$.

68. A bag holds 12 marbles, some red and the rest blue. $P(\text{red}) = \frac{1}{4}$. How many blue marbles are in the bag?

69. Design a spinner with 8 equal sections, using only red and blue, so that $P(\text{blue}) = 75\%$.

70. A wheel has the numbers 1 to 40 marked on it. When the wheel is spun, each number is equally likely to win.

(a) List all the multiples of 6 on the wheel. (1 mark)

(b) Find the probability that the winning number is a multiple of 6. Give your answer in simplest form. (1 mark)

(c) Find the probability that the winning number is a multiple of 6 that is greater than 20, as a percentage. (2 marks)

Revision

71. A letter is chosen at random from the word TOMORROW.

(a) List the sample space. (1 mark)

(b) How many outcomes are in the sample space? (1 mark)

(c) Find $P(\text{O})$ as a fraction, a decimal and a percentage. (2 marks)

(d) Describe this chance in words. (1 mark)

72. A map has a scale of $1 : 25\,000$.

(a) Two landmarks are 8 cm apart on the map. Find the actual distance in km. (2 marks)

(b) A walking track is 3.5 km long. Find its length on the map in cm. (2 marks)

(c) Kai walks the track in 50 minutes. Find his average speed in km/h. (1 mark)

73. A phone plan costs \$45 per month. GST is included in the price.

(a) Find the cost of the plan for one year. (1 mark)

(b) The price rises by 8%. Find the new monthly price. (2 marks)

(c) Find the GST in the new monthly price, to the nearest cent. (2 marks)

Solutions

1. 0.4, 75% and $\frac{2}{3}$. The others are less than 0 or more than 1.

2. (a) Impossible. (b) Even chance. (c) Likely. (d) Certain.

3. $0.9 \times 100 = 90\%$. Likely.

4. $0.02 \times 100 = 2\%$. Unlikely.

5. $2 \div 5 = 0.4 = 40\%$. Unlikely.

6. $17 \div 20 = 0.85 = 85\%$. Likely.

7. $1 \div 8 = 0.125 = 12.5\%$. Unlikely.

8. $\frac{60}{100} = \frac{3}{5}$, 0.6. Likely.

9. $\frac{4}{100} = \frac{1}{25}$, 0.04. Unlikely.

10. $\frac{72}{100} = \frac{18}{25}$, 0.72. Likely.

11. 0.3, 0.35, 0.4, so 0.3, 35%, $\frac{2}{5}$.

12. $\frac{7}{8} = 0.875$. In order: 0.85, $\frac{7}{8}$, 88%.

13. Lions: $9 \div 20 = 0.45$. Owls: 0.44. The Owls are least likely. The Hawks and Lions are equally likely (0.45 each).

14. (a) Even chance. (b) Impossible. (c) Certain.

15. C. Ella is 1 of 30 students, so it could happen but the chance is small.

16. Likely. Most of the counters (9 out of 10) are green.

17. A probability is between 0 and 1. 1.4 is more than 1, which would be more likely than certain.

18. $0.08 = 8\%$ and $0.5 = 50\%$. Since $8\% < 50\%$, 0.5 is more likely. He compared the digits, not the values.

19. $\frac{1}{2} = 0.5 = 50\%$

20. 0% means rain is impossible. A forecast is only an estimate, often rounded, so rain can still happen.

21. (a) $0.03 \times 100 = 3\%$. (b) Unlikely. (c) $\frac{1}{25} = 0.04 = 4\%$. Since $4\% > 3\%$, Factory B is more likely to make a faulty phone.

22. (a) $3 + 2 = 5$ parts, so $\frac{2}{5}$. (b) One part $= 4 \div 5 = 0.8$ L. Juice $= 3 \times 0.8 = 2.4$ L. (c) $\$3.60 \div 2 = \$1.80$ per litre. $2.4 \times \$1.80 = \$4.32$.

23. (a) $48 \div 60 = 0.8$, so 3.8 h. (b) $342 \div 3.8 = 90$ km/h. (c) $90\,000 \div 3600 = 25$ m/s. (d) $7.5 \times 3.42 = 25.65$ L. $25.65 \times \$1.96 = \$50.274 \approx \$50.27$.

24. (a) $72 \div 8 = 9$. (b) $\frac{8+9}{2} = 8.5$. (c) $15 - 4 = 11$. (d) Q_1 is the median of $4, 7, 7, 8$, so $Q_1 = 7$. Q_3 is the median of $9, 10, 12, 15$, so $Q_3 = 11$. IQR $= 11 - 7 = 4$.

25. $S = \{H, T\}$, 2 outcomes.

26. $S = \{A, B, C\}$, 3 outcomes.

27. $S = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\}$, 7 outcomes.

28. $S = \{G, Y\}$, 2 outcomes.

29. $S = \{R, B, W\}$, 3 outcomes.

30. $S = \{1, 2, 3\}$, 3 outcomes.

31. $S = \{M, I, S, P\}$, 4 outcomes.

32. $S = \{B, A, L, O, N\}$, 5 outcomes.

33. $S = \{S, T, A, I, C\}$, 5 outcomes.

34. 52

35. 28

36. $499 - 100 + 1 = 400$

37. 5, 6

38. 7, 14, 21, 28

39. 13, 15, 17, 19

40. The sample space lists the possible outcomes, each once. The spinner can only land on red, green or white, so there are 3 outcomes.

41. O is listed twice. Each outcome is listed once: $S = \{B, O, K\}$, so 3 outcomes.

42. Yes. Some numbers can appear on more than one section, as long as each of 1 to 5 appears at least once and no other number appears.

43. $S = \{\text{win, draw, loss}\}$, 3 outcomes.

44. (a) 30. (b) 22, 24, 26, 28, 30. (c) $30 - 5 = 25$. *cf. 2024 Std 1 HSC Q17*

45. (a) $3 \div 40 = 0.075$. (b) 7.5%. (c) Unlikely. (d) $7.5\% < 12\%$, so a car is more likely to be a ute.

46. (a) $38 \times \$28.40 = \$1079.20$. (b) $1.5 \times \$28.40 = \$42.60$ per hour. (c) $4 \times \$42.60 = \$170.40$. Total $= \$1079.20 + \$170.40 = \$1249.60$. (d) $2 \times \$1249.60 = \$2499.20$.

47. (a) $\$14.50 \div 5 = \$2.90$ per kg. (b) The 5 kg bag (\$2.90/kg is less than \$3.20/kg). (c) $5 \times \$3.20 = \$16.00$. Saving $= \$16.00 - \$14.50 = \$1.50$. (d) $5000 \div 150 = 33.3\ldots$, so 33 full serves.

48. $\frac{1}{2}$

49. 1, 3, 5: $\frac{3}{6} = \frac{1}{2}$

50. 1, 2, 3, 4: $\frac{4}{6} = \frac{2}{3}$

51. 4 green sections out of 6: $\frac{4}{6} = \frac{2}{3}$

52. $\frac{3}{10}$

53. $\frac{2}{12} = \frac{1}{6}$

54. Total $= 10$. $\frac{4}{10} = \frac{2}{5}$

55. Total $= 20$. $\frac{5}{20} = \frac{1}{4}$

56. Total $= 30$. $\frac{18}{30} = \frac{3}{5} = 60\%$

57. 6 letters, 3 are A: $\frac{3}{6} = \frac{1}{2}$

58. 10 letters, 3 are T: $\frac{3}{10}$

59. 11 letters. Vowels: A, E, A, I, so 4. $\frac{4}{11}$

60. 6, 8, 10: $\frac{3}{10}$

61. 4, 8, 12, 16, 20, 24, 28 and 6, 18, 30 (12 and 24 already counted): 10 numbers. $\frac{10}{30} = \frac{1}{3}$

62. 5, 15, 25, 35, 45: $\frac{5}{50} = 0.1$

63. 3 faces show 3: $\frac{3}{6} = \frac{1}{2}$

64. $\frac{4}{6} = \frac{2}{3}$

65. $P(5) = \frac{3}{6}$. Even: 2, 4, so $P(\text{even}) = \frac{2}{6}$. Rolling a 5 is more likely.

66. The colours are not equally likely: red covers half the spinner. $P(\text{red}) = \frac{1}{2}$.

67. The three results are not equally likely. Draws are much rarer than wins or losses.

68. Red $= \frac{1}{4} \times 12 = 3$. Blue $= 12 - 3 = 9$.

69. 75% of $8 = 6$. Six blue sections and two red sections.

70. (a) 6, 12, 18, 24, 30, 36. (b) $\frac{6}{40} = \frac{3}{20}$. (c) 24, 30, 36: $\frac{3}{40} = 7.5\%$. *cf. 2024 Std 1 HSC Q17*

71. (a) $S = \{T, O, M, R, W\}$. (b) 5. (c) 8 letters, 3 are O: $\frac{3}{8} = 0.375 = 37.5\%$. (d) Unlikely.

72. (a) $8 \times 25\,000 = 200\,000$ cm $= 2$ km. (b) 3.5 km $= 350\,000$ cm. $350\,000 \div 25\,000 = 14$ cm. (c) $3.5 \div \frac{50}{60} = 4.2$ km/h.

73. (a) $12 \times \$45 = \$540$. (b) $\$45 \times 1.08 = \$48.60$. (c) GST $= \$48.60 \div 11 = \$4.418\ldots \approx \$4.42$.